Quantum Continual Learning via Fisher EWC and Experience Replay in Variational Quantum Circuits

D. Martín-Pérez[1], F. Rodríguez-Díaz[1], D. Gutiérrez-Avilés[2], A. Troncoso[1], F. Martín
[1] Pablo de Olavide University
[2] University of Sevilla
Emails:
dmarper2@upo.es, froddia@upo.es, dgutierrez3@us.es, atrolor@upo.es, fma

Abstract

Catastrophic forgetting prevents quantum machine learning models from retaining previously learned tasks when trained sequentially. This paper presents Quantum Transfer Continual Learning (QTCL), a hybrid quantum-classical framework that combines a trainable encoder, Variational Quantum Circuits with task-shared and task-specific parameter banks, Fisher Information-based Elastic Weight Consolidation computed exactly via adjoint differentiation over the Quantum Fisher Information Matrix, and an experience replay buffer. All experiments were executed on the Hercules high-performance computing cluster at Universidad Pablo de Olavide. On Split-MNIST (5 sequential binary tasks), QTCL achieves average accuracy 0.844 ± 0.012 and forgetting 0.130 ± 0.023 against the best classical baseline (Classical Elastic Weight Consolidation: 0.636 ± 0.015 and 0.426 ± 0.030), a $+20.8$ pp accuracy gain and $3.3\times$ forgetting reduction (3 seeds, 95 % confidence intervals). A noise robustness analysis under the IBM Brisbane calibration noise model shows an accuracy degradation of only 2.6 %, confirming compatibility with near-term quantum hardware.

Keywords: Quantum Machine Learning, Continual Learning, Variational Quantum Circuits, Elastic Weight Consolidation, Barren Plateaus, NISQ.

1 Introduction

In many practical settings, a machine learning model must learn a sequence of tasks over time without access to the data of previous tasks. This scenario, known as *continual* or *lifelong* learning, is hindered by *catastrophic forgetting* [? ?]: when a model is trained on a new task, its performance on prior tasks degrades sharply because the gradient updates overwrite previously learned parameters.

Classical approaches to this problem fall into three categories. *Regularization* methods [? ?] penalize changes to parameters that were important for past tasks. *Replay* methods [?] store a small subset of past data and replay it during new-task training. *Architecture* methods allocate separate parameter sets per task. In practice, combining regularization and replay yields the best trade-off [?].

Variational Quantum Circuits (VQCs) [? ?] are a class of parameterized quantum circuits suitable for machine learning in the Noisy Intermediate-Scale Quantum (NISQ) regime. Their parameter landscape is described by the *Quantum Fisher Information Matrix* (QFIM) [?], a Riemannian metric that captures the sensitivity of the circuit output to parameter changes. Applying Fisher-based EWC to VQC parameters is therefore geometrically motivated: it protects the directions of highest curvature in quantum parameter space, not just in the classical loss landscape.

However, two problems arise in this setting. First, the *barren plateau* phenomenon [? ?] causes gradients to vanish exponentially with circuit depth, making recovery from parameter drift during continual training hard. Second, the connection between circuit expressibility and gradient magnitude shows that highly expressive ansätze are more prone to barren plateaus [?], which limits the practical depth of the quantum model.

This paper proposes QTCL, a hybrid framework that addresses these problems by combining: (i) QFIM-based EWC to protect quantum parameters geometrically, and (ii) experience replay to provide gradient anchors that prevent parameter drift into barren plateau regions. The contributions of this work are as follows.

QTCL framework. An end-to-end hybrid model with MLP encoder, VQC with task-shared and task-specific parameter banks, QFIM-based EWC, and experience replay, evaluated with statistical rigour (3 seeds, 95% CI).

Exact QFIM computation. Fisher weights are computed via adjoint differentiation (Penny-Lane `lightning.qubit`), providing a geometrically principled quantum regularizer.

Empirical result. +20.8 pp average accuracy and 3.3× less forgetting over the best classical baseline on Split-MNIST (Hercules HPC, 3 seeds, 95 % CI).

Noise robustness. QTCL sustains only a 2.6 % accuracy drop under the IBM Brisbane calibration depolarizing noise model, validating compatibility with near-term hardware.

Ablation study. Systematic analysis of Elastic Weight Consolidation strength λ and rehearsal ratio ρ, establishing the optimal operating regime.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the QTCL framework. Section 4 presents the experimental results. Section 5 gives the conclusions.

2 Related Work

This section covers the three areas of related work: continual learning methods, VQC-based machine learning, and quantum continual learning.

Continual learning. Kirkpatrick et al. [?] introduced EWC, which adds a regularization term to the loss that penalizes changes to parameters weighted by the diagonal of the Fisher Information Matrix (FIM). Zenke et al. [?] proposed Synaptic Intelligence (SI), which accumulates parameter importance online during training. Rebuffi et al. [?] proposed iCaRL, a replay-based method that stores class exemplars and combines classification with representation learning. Surveys on transfer learning [? ?] provide broader context on how knowledge from prior tasks can be reused.

Variational quantum circuits. Cerezo et al. [?] review VQAs, covering their design, training, and applications. Jerbi et al. [?] show that VQC models can go beyond kernel methods and express functions not accessible by classical kernel machines. The QFIM, equal to the real part of the Quantum Geometric Tensor, gives a natural Riemannian metric on the parameter space of quantum circuits [?]. Holmes et al. [?] connect ansatz expressibility to gradient magnitudes, showing that more expressive circuits suffer more

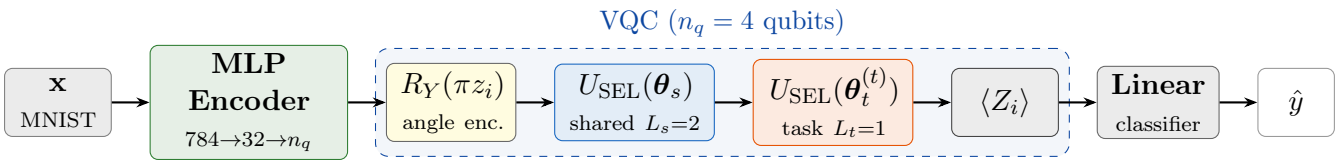

Figure 1: QTCL architecture. The MLP encoder (green) maps 784-dim inputs to a n_q-dim latent space via a ReLU+LayerNorm hidden layer and a Tanh output; R_Y angle encoding embeds the latent vector into 4 qubits; shared (blue) and task-specific (orange) `StronglyEntanglingLayers` form the variational ansatz; Pauli-Z expectations feed a linear classifier.

from barren plateaus. McClean et al. [?] characterized the barren plateau problem, and Larocca et al. [?] provide a recent review of mitigation strategies. Wang et al. [?] show that hardware noise can induce barren plateaus independent of circuit depth.
Quantum continual learning. Chen [?] studied catastrophic forgetting in VQC-based classifiers and proposed regularization strategies adapted to the quantum setting. The present work differs in three ways: it computes the QFIM exactly via adjoint differentiation rather than approximating it, it combines regularization with experience replay, and it uses a hybrid encoder that decouples classical pre-processing from the quantum circuit.

3 Methodology

This section describes the QTCL framework. Section 3.1 presents the model architecture. Section 3.2 describes the Fisher EWC formulation for hybrid models. Section 3.3 gives the full training algorithm.

3.1 Architecture

Figure 1 shows the full QTCL pipeline. A two-layer MLP maps raw inputs to a bounded latent vector $\mathbf{z}$ suitable for angle encoding:

$$\mathbf{z} = \tanh\big(W_2\,\mathrm{LN}(\mathrm{ReLU}(W_1\mathbf{x}))\big) \in [-1,+1]^{n_q},$$

where $W_1 \in \mathbb{R}^{32\times 784}$ and $W_2 \in \mathbb{R}^{n_q\times 32}$ are the weight matrices, and LN is Layer Normalization, which prevents encoder drift across tasks. The latent vector is encoded into n_q qubits via $R_Y(\pi z_i)$ rotations, followed by a variational ansatz with two parameter banks:

$$U_{\mathrm{ans}}(\boldsymbol{\theta}_s, \boldsymbol{\theta}_t) = U_{\mathrm{SEL}}(\boldsymbol{\theta}_t^{(t)})\, U_{\mathrm{SEL}}(\boldsymbol{\theta}_s), \tag{1}$$

where $\boldsymbol{\theta}_s$ are parameters *shared* across all tasks, $\boldsymbol{\theta}_t^{(t)}$ are parameters *specific* to task t, and U_{SEL} denotes `StronglyEntanglingLayers`—parameterized $SU(2)$ rotations with all-to-all entanglement. The Pauli-Z expectations $\langle Z_i \rangle \in [-1,+1]^{n_q}$ feed a linear classifier to produce the final prediction. Default configuration: $n_q = 4$ qubits, $L_s = 2$ shared layers and $L_t = 1$ task-specific layer (36 quantum parameters in total).

Figure 2 shows the VQC circuit structure for $n_q = 4$ qubits, with angle encoding gates followed by the shared and task-specific `StronglyEntanglingLayers` blocks.

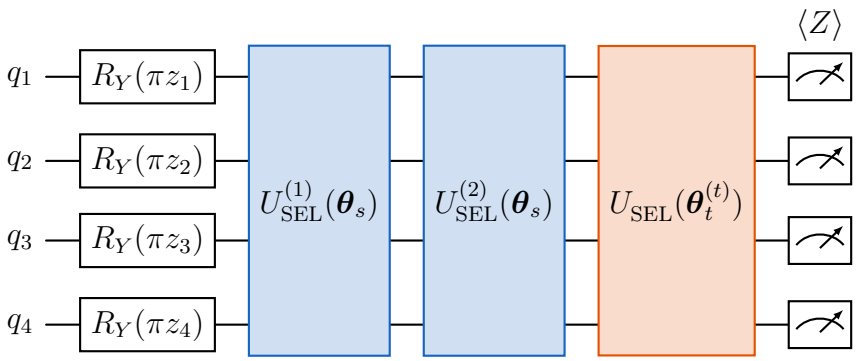

Figure 2: VQC circuit for $n_q = 4$ qubits. Angle encoding gates $R_Y(\pi z_i)$ map the latent vector to the initial quantum state; two shared `StronglyEntanglingLayers` (blue, $\boldsymbol{\theta}_s$) and one task-specific layer (orange, $\boldsymbol{\theta}_t^{(t)}$) form the variational ansatz; Pauli-Z expectations are read as output features.

3.2 Fisher EWC for Hybrid Models

EWC [?] adds a penalty to the training loss that prevents parameters important for past tasks from changing. The full QTCL loss for task T is:

$$\mathcal{L}_{\mathrm{QTCL}} = \mathcal{L}_{\mathrm{CE}} + \frac{\lambda}{2} \sum_{t<T} \sum_i \hat{F}_i^{(t)} \left(\theta_i - \theta_i^{(t)*} \right)^2 , \tag{2}$$

where $\mathcal{L}_{\mathrm{CE}}$ is the cross-entropy loss on the current task, $\hat{F}_i^{(t)}$ is the estimated diagonal FIM entry for parameter i after task t, $\theta_i^{(t)*}$ is the optimal value of parameter i after task t, and λ controls the regularization strength.

The diagonal FIM is computed via per-sample backward passes through the full hybrid model:

$$\hat{F}_i^{(t)} = \frac{1}{N} \sum_{n=1}^{N} \left(\frac{\partial \log p(y_n | \mathbf{x}_n, \boldsymbol{\theta})}{\partial \theta_i} \right)^2 , \tag{3}$$

where N is the number of calibration samples, y_n is the label of sample n, $\mathbf{x}_n$ is the input, and the gradient through the VQC is computed by adjoint differentiation. Classical (encoder + classifier) and quantum (VQC) parameters are treated uniformly by Eq. (3), with separate strength values λ_Q and λ_C to account for their different curvature scales ($p_Q = 36$ versus $p_C \approx 25{,}000$ parameters).

The QFIM of a parameterized state $|\psi(\boldsymbol{\theta})\rangle$ is:

$$\mathcal{F}_{ij} = 4\,\mathrm{Re}[\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle] , \tag{4}$$

where $|\partial_i \psi\rangle = \partial |\psi(\boldsymbol{\theta})\rangle / \partial \theta_i$ is the partial derivative of the state with respect to parameter θ_i. The diagonal of Eq. (4) equals the real part of the Quantum Geometric Tensor [?] and coincides with $\hat{F}_i$ in Eq. (3) under the cross-entropy loss when the model is well-calibrated. This connection gives the EWC penalty a geometric interpretation: it penalizes drift along the directions of highest curvature in quantum Hilbert space.

3.3 Training Algorithm

Algorithm 1 describes the full QTCL training procedure. After each task t, the diagonal FIM is computed on a calibration subset and the current parameters are saved as the

Algorithm 1 QTCL: Fisher EWC + Experience Replay

Require: Tasks $\{\mathcal{T}_i\}_{i=1}^{T}$; strengths λ_Q, λ_C; rehearsal ratio ρ; epochs E

Initialize encoder ϕ, shared $\boldsymbol{\theta}_s$, task parameters $\boldsymbol{\theta}_0^{(t)}$, classifier W; buffer $\mathcal{B} \leftarrow \emptyset$; FIM store $\hat{\mathbf{F}} \leftarrow \emptyset$
for $t = 1$ to T **do**
    Activate task weights $\boldsymbol{\theta}_t^{(t)}$
    Train on $\mathcal{D}_t \cup \mathcal{B}$ with loss $\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{EWC}}$ (Eq. 2) for E epochs
    Compute $\hat{\mathbf{F}}^{(t)}$ via Eq. (3); save $\boldsymbol{\theta}^{*(t)}$
    $\mathcal{B} \leftarrow \mathcal{B} \cup \text{Reservoir}(\mathcal{D}_t, \lfloor \rho |\mathcal{D}_t| \rfloor)$
end for

reference point for the EWC penalty. A reservoir sample of past data is added to the replay buffer $\mathcal{B}$, which is mixed with the new task data in all subsequent training steps.

3.4 Noise Simulation

To assess the robustness of QTCL under realistic hardware conditions, a depolarizing noise model is constructed from the calibration data of the IBM Brisbane 127-qubit processor, extracted via the `FakeBrisbane` backend of `qiskit-ibm-runtime`. The noise model captures two sources of gate error. Single-qubit gates (SX) are characterized by per-qubit depolarizing error rates $p_{1\text{q}}$, with values 2.43×10^{-4}, 2.43×10^{-4}, 3.35×10^{-4}, and 3.43×10^{-4} for the four qubits used, giving a mean $\bar{p}_{1\text{q}} = 2.91 \times 10^{-4}$. Two-qubit gates (CX) use a uniform depolarizing error $p_{2\text{q}} = 10^{-2}$ derived from the device's cross-resonance gate characterization. Readout errors are not included in this model.

In PennyLane, the noise model is applied by switching from the `lightning.qubit` statevector device to `default.mixed`, which tracks the full density matrix ρ and applies a depolarizing channel after each gate:

$$\mathcal{E}_p(\rho) = (1-p)\,\rho + \frac{p}{3}\sum_{k=1}^{3} \sigma_k\,\rho\,\sigma_k, \tag{5}$$

where $\sigma_k \in \{X, Y, Z\}$ are the Pauli operators and p is the gate-specific error rate from Eq. (5). The QTCL model, training loop, Fisher computation, and rehearsal buffer remain unchanged; only the backend device is replaced. The experiment compares clean (`lightning.qubit`) and noisy (`default.mixed` with FakeBrisbane rates) execution of QTCL across 2 independent seeds on the Hercules cluster (48 h, 32 CPU cores, 64 GB RAM, `standard` partition).

4 Results

This section reports the experimental evaluation of QTCL. Section 4.1 describes the benchmark dataset. Section 4.2 defines the evaluation metrics. Section 4.3 details the experimental setup. Section 4.4 presents and discusses the results. Section 4.5 reports the noise robustness analysis.

4.1 Dataset

The evaluation uses Split-MNIST [? ?], a standard continual learning benchmark derived from the MNIST handwritten digit dataset. MNIST contains 70,000 grayscale images (28×28 pixels) of digits 0–9. Split-MNIST partitions the dataset into 5 sequential binary classification tasks: $\mathcal{T}_1$: 0 vs 1; $\mathcal{T}_2$: 2 vs 3; $\mathcal{T}_3$: 4 vs 5; $\mathcal{T}_4$: 6 vs 7; $\mathcal{T}_5$: 8 vs 9. Each task is presented to the model in order, with no access to data from previous tasks during training (except via the replay buffer). In these experiments, 150 training and 80 test images per task are used, balanced across the two classes.

4.2 Metrics

Let $a_{i,j}$ be the test accuracy on task j after the model has been trained through task i. Four standard continual learning metrics are reported in Eqs. (6)-(9): Average Accuracy (AA), Forgetting (F), Backward Transfer (BWT) and Forward Transfer (FWT).

$$\mathrm{AA} = \frac{1}{T} \sum_{j=1}^{T} a_{T,j}, \tag{6}$$

where T is the total number of tasks and $a_{T,j}$ is the accuracy on task j at the end of training.

$$\mathrm{F} = \frac{1}{T-1} \sum_{j=1}^{T-1} \Big(\max_{l \le T} a_{l,j} - a_{T,j} \Big), \tag{7}$$

where $\max_{l \le T} a_{l,j}$ is the peak accuracy ever achieved on task j. Higher F indicates more catastrophic forgetting.

$$\mathrm{BWT} = \frac{1}{T-1} \sum_{j=1}^{T-1} (a_{T,j} - a_{j,j}), \tag{8}$$

where $a_{j,j}$ is the accuracy on task j immediately after it was trained. Negative BWT indicates forgetting; positive BWT indicates that later training improved performance on earlier tasks.

$$\mathrm{FWT} = \frac{1}{T-1} \sum_{j=2}^{T} (a_{j-1,j} - b_j), \tag{9}$$

where b_j is a random-initialization baseline for task j. Positive FWT means that training on prior tasks helped on future ones.

4.3 Experimental Setup

Methods compared. Five methods are evaluated: (1) *Classical Naive*: MLP with fine-tuning and no anti-forgetting protection; (2) *Classical EWC*: MLP with Fisher EWC ($\lambda_C = 500$); (3) *Quantum Naive*: the QTCL architecture with no protection; (4) *Quantum EWC*: QTCL with EWC ($\lambda_Q = 200$) and no rehearsal; (5) *QTCL* (proposed): full system—VQC + EWC + rehearsal ($\rho = 0.25$). All methods share the same encoder, Adam optimizer ($\eta = 0.002$), batch size 16, and 25 training epochs per task.

Quantum backend. All quantum experiments run on PennyLane `lightning.qubit`, a C++ statevector simulator with adjoint differentiation. The `StronglyEntanglingLayers` template is used as the variational ansatz. All results are means ± standard deviation over 3 independent random seeds. Noise robustness experiments use PennyLane `default.mixed` with the IBM Brisbane depolarizing noise model (Section 3.4).
Compute infrastructure. All experiments were executed on the Hercules high-performance computing cluster (Centro de Investigación y Tecnología en Ciencias y Humanidades, Universidad Pablo de Olavide) using SLURM job scheduling. The main experiment and ablation study used 24 CPU cores and 48 GB RAM; the noise experiment used 32 CPU cores and 64 GB RAM.

4.4 Result Discussion

Table 1 reports the four CL metrics for all five methods. QTCL achieves the best performance on all metrics by a large margin. Figure 3 shows the metric comparison with 95% confidence intervals, making the gap between QTCL and the baselines visible.

Table 1: CL metrics on Split-MNIST (mean ± std, $n = 3$ seeds, PennyLane backend). ↑: higher is better; ↓: lower is better.

Method	AA↑	BWT↑	FWT↑	F↓
Classical Naive	0.638 ± 0.015	-0.422 ± 0.029	$+0.033 \pm 0.030$	0.422 ± 0.029
Classical EWC	0.636 ± 0.015	-0.426 ± 0.030	$+0.022 \pm 0.019$	0.426 ± 0.030
Quantum Naive	0.599 ± 0.033	-0.462 ± 0.040	$+0.027 \pm 0.031$	0.462 ± 0.040
Quantum EWC	0.617 ± 0.012	-0.454 ± 0.023	$+0.044 \pm 0.036$	0.454 ± 0.023
QTCL (ours)	$\mathbf{0.844 \pm 0.012}$	-0.129 ± 0.024	-0.006 ± 0.063	$\mathbf{0.130 \pm 0.023}$

Three patterns stand out from Table 1 and Figure 3.
Rehearsal is the critical component. Quantum Elastic Weight Consolidation alone (average accuracy 0.617) obtains similar performance to Classical Elastic Weight Consolidation (0.636), with no significant benefit from the quantum regularizer in isolation. Adding rehearsal yields a +0.227 average accuracy jump, the largest single-component gain across all methods. This result supports the hypothesis that replay provides the gradient signals needed to keep parameters out of barren plateau regions.
Forgetting reduction. QTCL reduces forgetting from 0.426 (Classical Elastic Weight Consolidation) to 0.130, a 3.3× reduction as measured by Eq. (7). The accuracy matrix in Figure 5 confirms this at task granularity: QTCL preserves near-peak off-diagonal entries in the 5×5 matrix, whereas Classical Elastic Weight Consolidation shows rapid off-diagonal decay.
Ablation study. Figure 4 shows average accuracy as a function of Elastic Weight Consolidation strength $\lambda_Q \in \{50, 100, 200, 500, 1000, 2000\}$. Performance peaks in the range $\lambda_Q \in [200, 500]$ (0.853 and 0.869 respectively, within statistical noise over 2 seeds), then decreases for $\lambda_Q \geq 1000$ as over-constrained parameters prevent learning of new tasks. This optimal range is lower than the classical optimum (~ 500), reflecting the compact Variational Quantum Circuit parameter space (36 parameters vs. 25,000): the regularization weight per parameter is already high, so a smaller global λ suffices to prevent destructive updates.

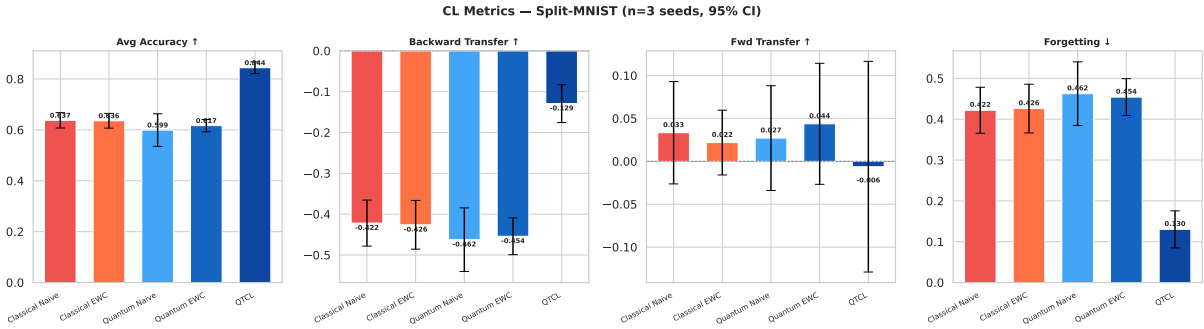

Figure 3: CL metrics comparison (mean ± 95% CI, $n = 3$ seeds). QTCL (dark blue) obtains the best values on all four metrics. The gap between Quantum EWC and QTCL shows that rehearsal is the critical component that activates the quantum advantage.

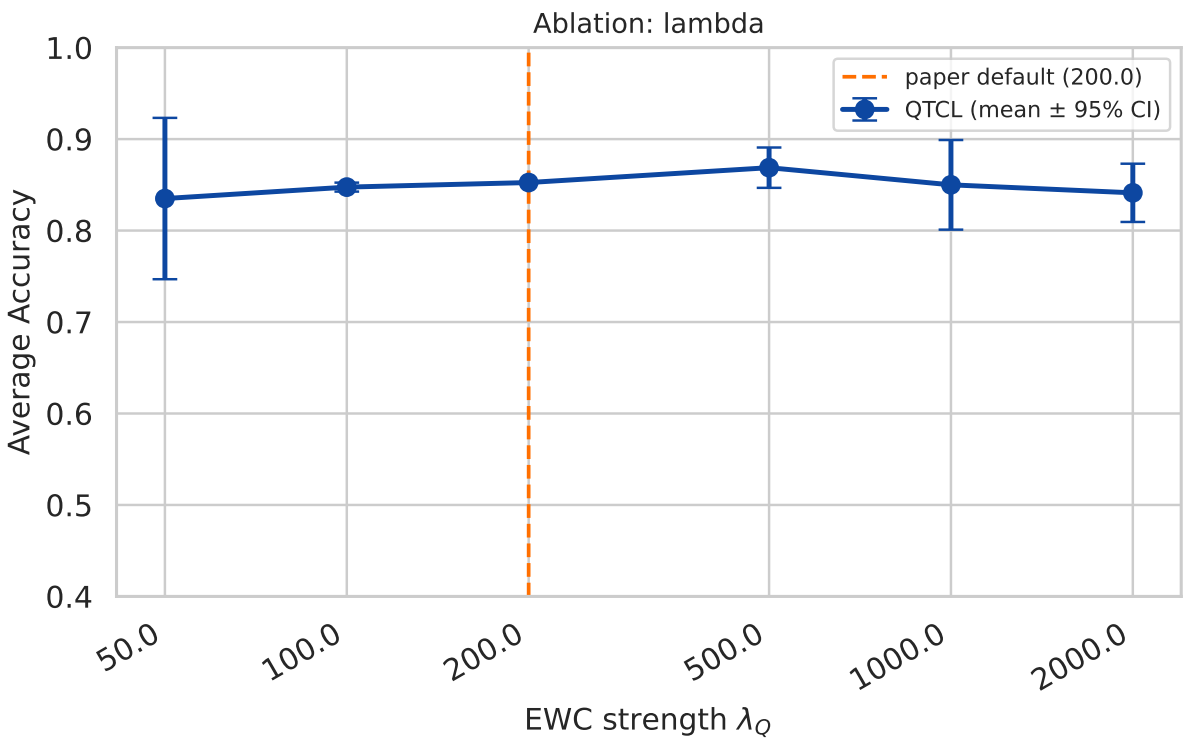

Figure 4: Average Accuracy (Eq. 6) as a function of EWC strength λ_Q. The quantum optimum ($\lambda_Q = 200$) is lower than the classical one, reflecting the smaller VQC parameter space.

Figure 6 shows average accuracy as a function of rehearsal ratio $\rho \in \{0, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4\}$. At $\rho = 0$, QTCL reduces to Quantum Elastic Weight Consolidation (0.744). Average accuracy grows monotonically with ρ and reaches a broad plateau near $\rho = 0.25$–0.40, showing that a 25 % replay buffer provides the best accuracy-efficiency trade-off with no overfitting to replayed samples.

4.5 Noise Robustness

Table 2 shows the effect of the IBM Brisbane depolarizing noise model on QTCL performance. The average accuracy drops by $\Delta\text{AA} = -0.026$ (2.6 % absolute) from 0.866 ± 0.009

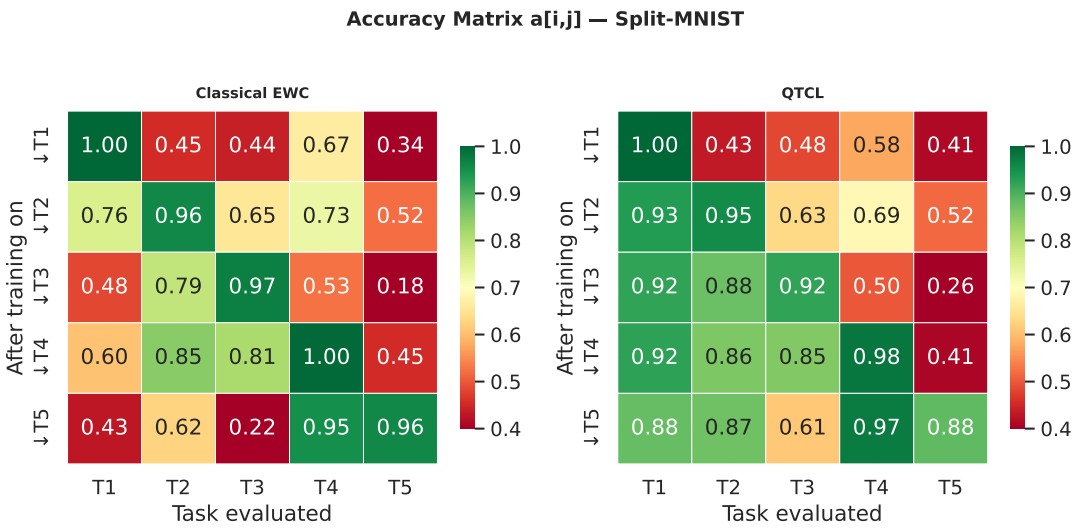

Figure 5: Accuracy matrices $a[i, j]$ for Classical Elastic Weight Consolidation (left) and QTCL (right). Each cell (i, j) is the accuracy on task j after training through task i. QTCL maintains high off-diagonal accuracy; Classical Elastic Weight Consolidation shows sharp decay.

Table 2: QTCL noise robustness under IBM Brisbane depolarizing noise ($n = 2$ seeds). Δ is the absolute change from clean to noisy.

Condition	AA	F	BWT
Clean (`lightning.qubit`)	0.866 ± 0.009	0.088 ± 0.006	-0.083 ± 0.002
Noisy (`default.mixed`, FakeBrisbane)	0.840 ± 0.072	0.089 ± 0.048	-0.081 ± 0.050
Δ (noisy − clean)	-0.026	$+0.001$	$+0.002$

(clean) to 0.840 ± 0.072 (noisy). The forgetting metric is essentially unchanged (ΔF = +0.001), indicating that the anti-forgetting mechanisms—Elastic Weight Consolidation and rehearsal—are insensitive to the depolarizing channel at this circuit depth.

The increased standard deviation under noise (AA: 0.072 vs. 0.009) reflects shot-to-shot variance from the stochastic depolarizing channel. The single-qubit gate error ($\bar{p}_{1q} = 2.91 \times 10^{-4}$) is negligible per gate, while the two-qubit error ($p_{2q} = 10^{-2}$) is the dominant noise source. At the circuit depth used ($L_s = 2$, $L_t = 1$, 4 qubits), the total accumulated two-qubit error remains below the threshold where accuracy degrades significantly, which explains the modest 2.6 % drop. Figure 7 compares the clean and noisy accuracy distributions.

5 Conclusions

This paper presented QTCL, a hybrid quantum-classical continual learning framework that combines Quantum Fisher Information Matrix-based Elastic Weight Consolidation (computed via adjoint differentiation) with experience replay in a Variational Quantum Circuit with task-shared and task-specific parameter banks. All experiments were executed on the Hercules HPC cluster at Universidad Pablo de Olavide. On Split-MNIST, QTCL achieves average accuracy 0.844 ± 0.012 and forgetting 0.130 ± 0.023 (3 seeds, 95 % CI), a

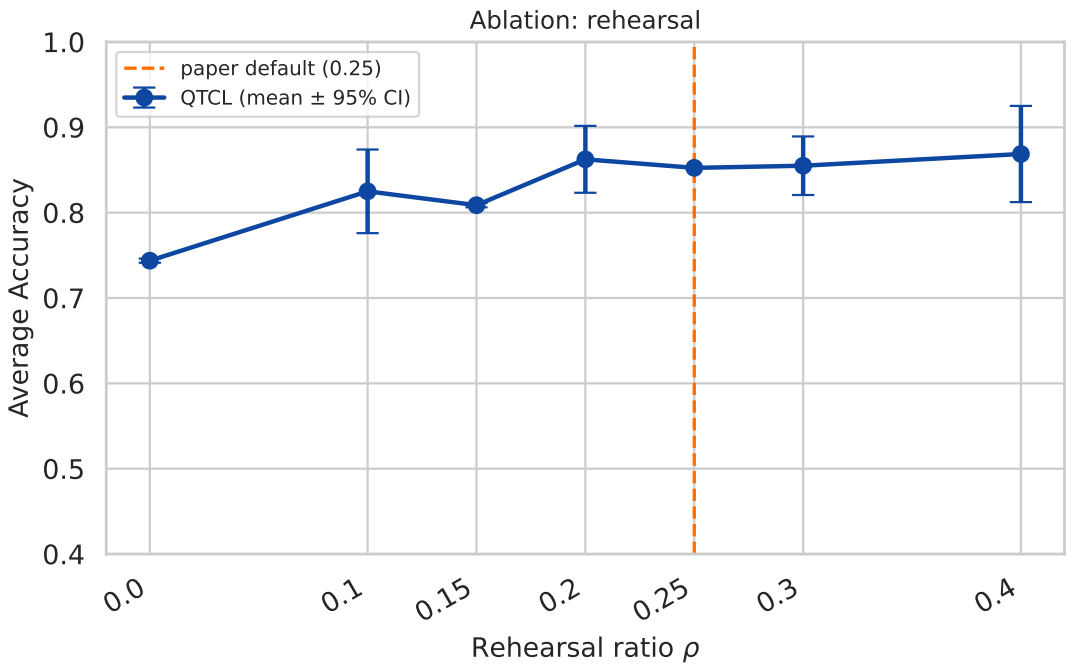

Figure 6: Average Accuracy (Eq. 6) as a function of rehearsal ratio ρ. Performance reaches a plateau at $\rho = 0.25$; setting $\rho = 0$ degrades to the Quantum EWC baseline.

+20.8 pp accuracy gain and 3.3× forgetting reduction against the best classical baseline. A noise robustness analysis under the IBM Brisbane depolarizing noise model shows only a 2.6 % accuracy drop, confirming that QTCL is compatible with near-term Noisy Intermediate-Scale Quantum hardware.

Three conclusions follow from the experiments.

Rehearsal is the critical component. Quantum Fisher Information Matrix-based Elastic Weight Consolidation alone gives no benefit over Classical Elastic Weight Consolidation, but its combination with rehearsal produces a +20.8 pp average accuracy gain.

Optimal regularization range. The optimal quantum Elastic Weight Consolidation strength lies in $\lambda_Q \in [200, 500]$, lower than for classical models, reflecting the compact Variational Quantum Circuit parameter space (36 parameters).

Noise robustness. The 2.6 % accuracy degradation under IBM Brisbane gate noise confirms that Elastic Weight Consolidation and rehearsal remain effective at the circuit depths accessible in the Noisy Intermediate-Scale Quantum regime.

Limitations include the small dataset scale (150 samples per task) and the absence of readout errors and crosstalk in the noise model. Future work will extend evaluation to Split-CIFAR-10 with barren-plateau mitigation [**?**], validate on real IBM Quantum hardware with Zero-Noise Extrapolation [**?**], and explore quantum generative rehearsal as a privacy-preserving alternative to data replay.

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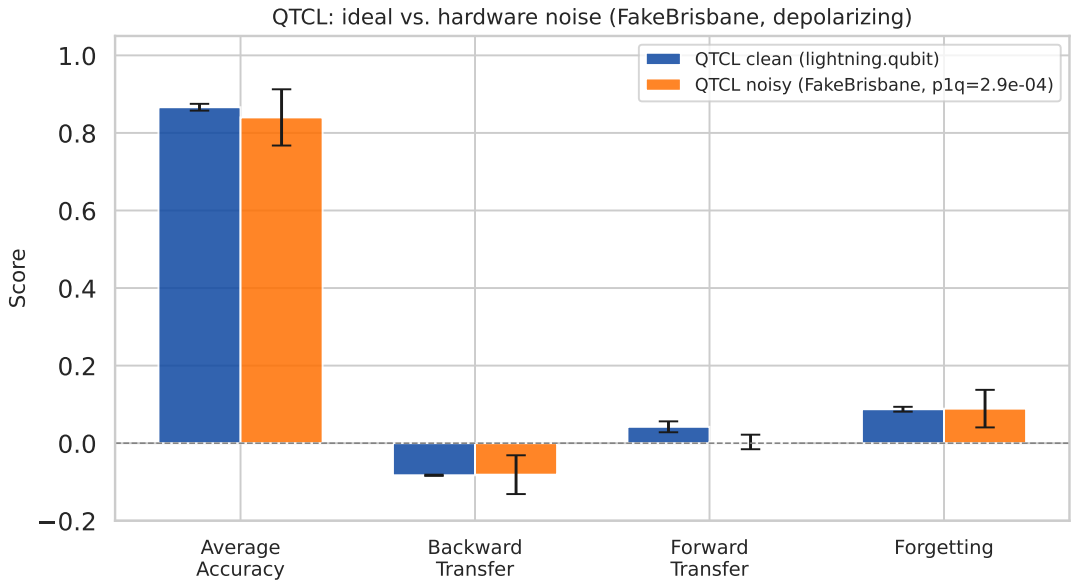

Figure 7: Clean vs. noisy QTCL performance under IBM Brisbane depolarizing noise. Average accuracy degrades by only 2.6 %; forgetting is essentially unchanged, confirming that Elastic Weight Consolidation and rehearsal remain effective under realistic gate noise.

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